SMART INDIA HACKATHON 2025

TITLE PAGE

- **Problem Statement ID – 25050**
- **Problem Statement Title- Smart Traffic Management System for Urban Congestion**
- **Theme- Transportation & Logistics**
- **PS Category- Software**
- **Team ID-**
- **Team Name (Registered on portal) - StuckInTraffic**

SwiftSignals

•"A smart traffic signal system that uses existing cameras and predictive algorithms to adapt signal timings in real time, reducing congestion and wait times." 🚦

•Uses **existing traffic cameras + computer vision** to monitor vehicles.
•Computes a **traffic index (c)** based on vehicle count vs road capacity.
•Determines **green signal time** using vehicle discharge rate.
•Ensures **fair signal allocation between lanes**.
•Predicts **incoming traffic from nearby signals** using travel time and past data.
•**Recalculates traffic before each cycle** for dynamic control.
•**Reduces congestion, waiting time, and fuel consumption** without extra hardware.

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TECHNICAL APPROACH

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We define a Traffic Index (c): $0 < c < 1$

$N = c \cdot N_{max}$

Where: N_{max} = vehicle count indicating full congestion.

Fraction of Traffic to Clear

$f(c) = 1$, if $c < 0.4$

1 - 1.5(c-0.4)) , if $0.4 \leq c \leq 0.6$

0.7 , if $c > 0.6$

$$t = \frac{N_{clear}}{R}$$

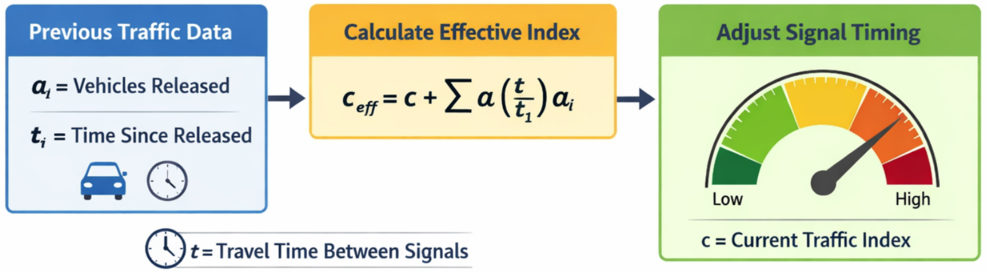
Signal is also time bounded by t1 and t2.

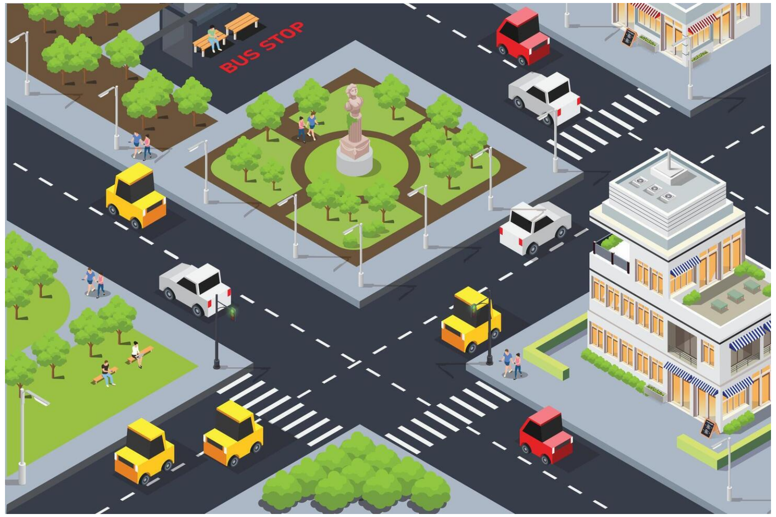
Programming & Technologies

- Python – core implementation language
- Computer Vision (OpenCV / Deep Learning models) – vehicle detection and counting from traffic cameras
- Machine Learning – adaptive analysis of traffic patterns and system improvement over time

Effective Traffic Index

$$c_{eff} = c_0 + \sum \alpha \left(\frac{t_i}{t_1}\right) a_i$$





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FEASIBILITY AND VIABILITY

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Feasibility

•Uses existing traffic cameras, avoiding new hardware costs.
•Computer vision models detect and count vehicles in real time.
•Can integrate with programmable traffic controllers.
•Runs on central servers or edge devices, enabling city-scale deployment.

Challenges

•Driver behavior issues (red-light jumping, lane violations).
•Seasonal traffic variations affecting travel time.
•Unexpected spikes from accidents, events, or construction.
•Poor camera visibility in rain, fog, or night.

Solutions

•Train AI/ML models on diverse traffic data.
•Continuously retrain models to adapt to seasonal changes.
•Use dynamic parameter updates for variables like travel time.
•Combine historical + real-time data to detect abnormal congestion quickly.

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IMPACT AND BENEFITS

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Impact

- Commuters: Shorter waits and smoother travel.
- Traffic Authorities: Data-driven traffic management.
- Public Transport & Emergency Services: Faster, more reliable routes.
- Urban Planners: Access to real-time and historical traffic data.

Benefits

- Social: Less commuter stress, improved road safety.
- Economic: Reduced fuel waste and productivity loss.
- Environmental: Lower emissions and better air quality.

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RESEARCH AND REFERENCES

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- https://www.tomtom.com/traffic-index/ranking
- https://idd.insaan.iiit.ac.in/dataset/details/
- https://www.roadvision.ai/blog/the-role-of-computer-vision-in-modern-traffic-management-systems